Plan for the rest of this week

- Today
  - Discuss Poisson exercise
  - Lectures on mixed models
  - Interactive session on statistics teaching (with Allyson and Emile)
  - Exercise on mixed models
  - (Work on Poisson exercise)
- Friday (Saturday)
  - Deadline for the second report

Processing and Analysis of Biological Data

BIOS15 2025





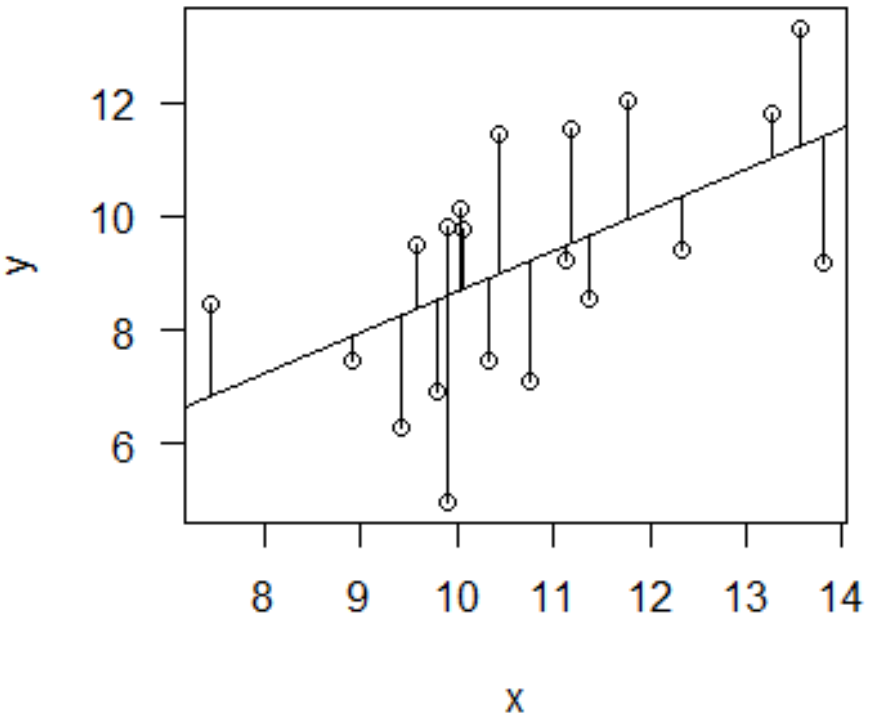
Lecture 7.
Linear Mixed Models

Masahito Tsuboi/Øystein H. Opedal

$$y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i$$

The linear model

- Most of the models we will work with in this course are linear models, that describe how a linear set of predictors relate to a response variable
- A key element of the model is the so-called linear predictor:
- $y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i, \varepsilon \sim N(0, \sigma^2)$
- The term $\varepsilon \sim N(0, \sigma^2)$ means that the residuals (epsilon) are assumed to follow a normal distribution



Linear mixed models

- Linear mixed models extend the linear model by introducing **random effects**, i.e. variables for which we are more interested in the variance than the mean
- Often used to account for hierarchical structure of sampling (e.g. individuals nested within different kinds of patches)
- Assumes normally distributed group means, and normally distributed residuals within groups

$$y = \mathbf{X}\beta + \mathbf{Z}\mu + \epsilon$$



Fixed or random effect?

Fixed effect:

- Interested in the specific levels
- Estimates a coefficient for each level
- Treat levels as “the entire set”

Random effect:

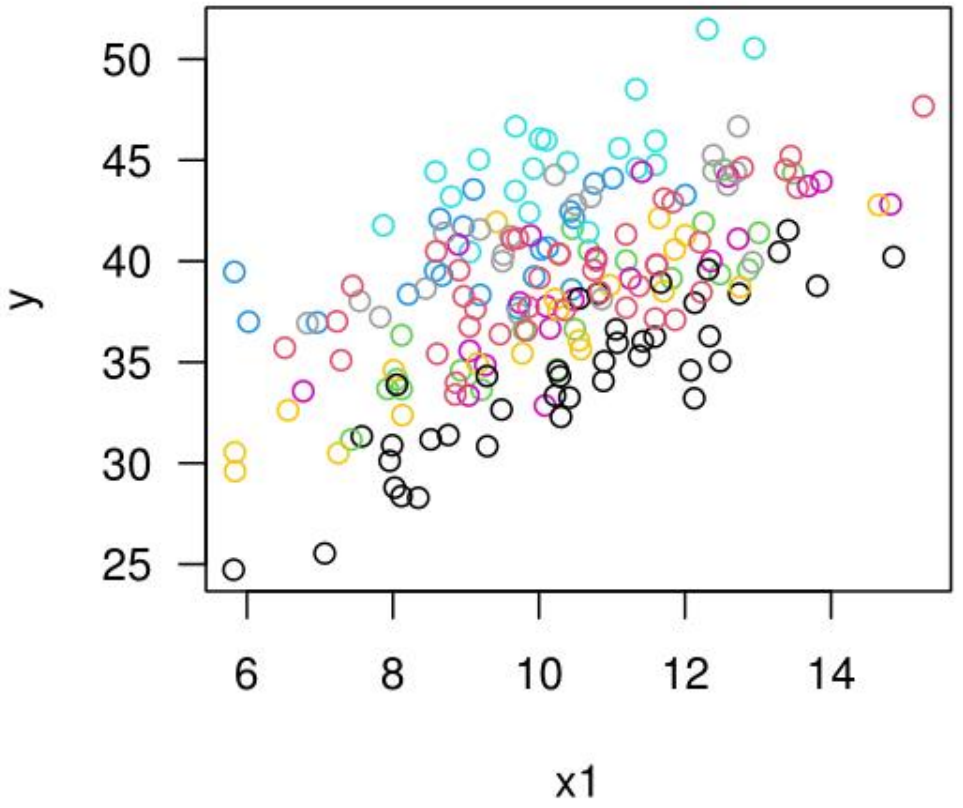
- Interested in variation among levels
- Estimates a variance component
- Treat levels as a sample from a broader population

Same data → different choice depending on the goal

Linear mixed models

$$y = \mathbf{X}\beta + \mathbf{Z}\mu + \epsilon$$

- Linear mixed models extend the linear model by introducing **random effects**, i.e. variables for which we are more interested in the variance than the mean
- Often used to account for hierarchical structure of sampling (e.g. plots nested in forest patches)
- Assumes normally distributed group means, and normally distributed residuals within groups



Variance component analysis

- We can use a model with only an intercept as fixed effect to partition variance across hierarchical levels
- Here, the % variance explained by groupID is 9.205/(9.205+11.869)
- The % variance within groups is 11.869/(9.205+11.869)
- The intercept is the mean of the response variable

```
##  Family: gaussian  ( identity )
## Formula:          y ~ 1 + (1 | groupID)
## Data: data
##
##      AIC      BIC   logLik deviance df.resid
##   1096.4   1106.3   -545.2   1090.4      197
##
## Random effects:
##
## Conditional model:
##  Groups   Name        Variance Std.Dev.
##  groupID  (Intercept)  9.205   3.034
##  Residual             11.869   3.445
## Number of obs: 200, groups:  groupID, 10
##
## Dispersion estimate for gaussian family (sigma^2): 11.9
##
## Conditional model:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  38.7679     0.9899   39.16   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Variance component analysis

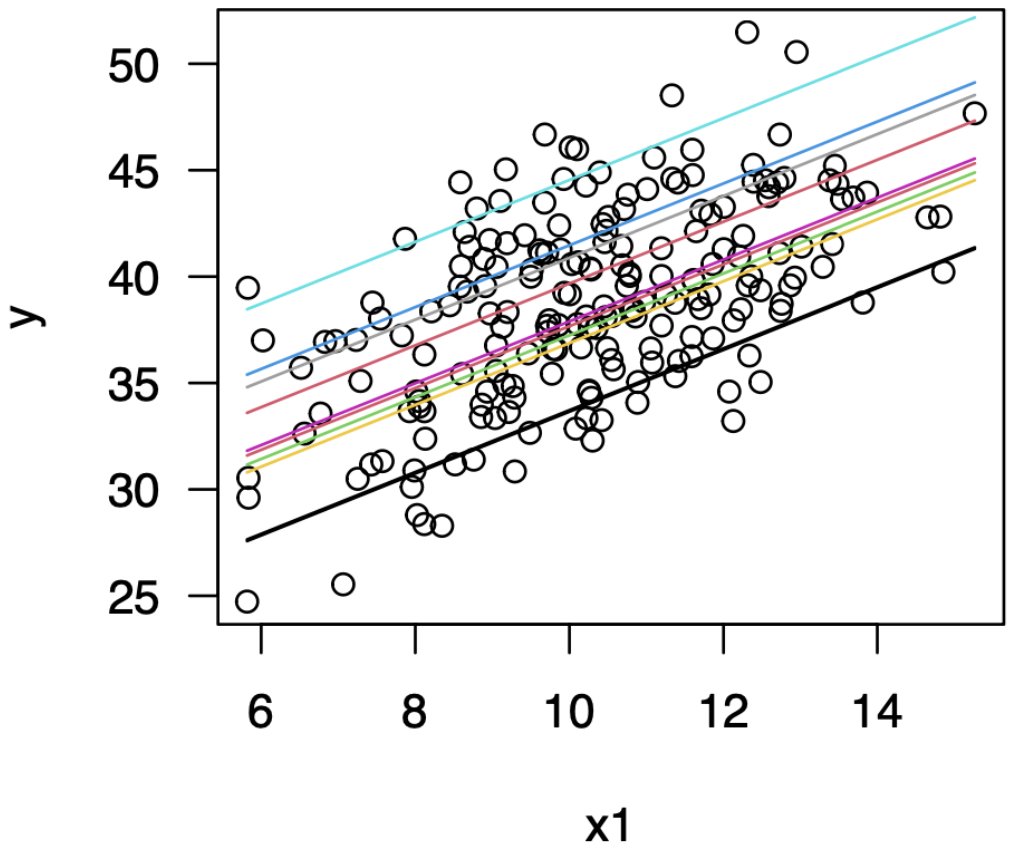
Table S4. Components of phenotypic variance in blossom traits

Trait	Mean ± SE	Among populations	Among families	Among plants	Within plants	CV^2 (%)
Field:						
UBA (mm^2)	373.62 ± 15.62	0.020 (0.004, 0.043)		0.020 (0.007, 0.035)	0.062 (0.048, 0.078)	10.2
		19.9 %		19.3 %	60.8 %	
GA (mm^2)	23.73 ± 1.05	0.024 (0.005, 0.053)		0.027 (0.014, 0.041)	0.036 (0.028, 0.047)	8.7
		28.1 %		30.6 %	41.3 %	
ASD (mm)	3.97 ± 0.27	0.072 (0.025, 0.138)		0.014 (0.004, 0.029)	0.061 (0.046, 0.083)	14.7
		48.6 %		9.9 %	41.5 %	
Greenhouse:						
UBA (mm^2)	623.79 ± 24.19	0.005 (0, 0.025)	0.045 (0.018, 0.075)	0.026 (0.010, 0.045)	0.032 (0.024, 0.042)	10.6
		5.0 %	41.3 %	24.0 %	29.6 %	
GA (mm^2)	25.48 ± 1.18	0.020 (0, 0.051)	0.017 (0.001, 0.037)	0.017 (0.004, 0.033)	0.036 (0.026, 0.048)	8.9
		22.3 %	18.9 %	18.9 %	39.9 %	
ASD (mm)	3.70 ± 0.31	0.083 (0.017, 0.187)	0.047 (0.019, 0.083)	0.007 (0, 0.023)	0.057 (0.037, 0.085)	19.4
		42.6 %	24.1 %	3.6 %	29.6 %	

Variance components with 95% confidence intervals are scaled by the square of the trait mean, and summed to obtain CV^2. The reported percentages are the percent of the total variance residing at each hierarchical level.

Random-intercept model

```
##  Family: gaussian  ( identity )
## Formula:          y ~ x1 + (1 | groupID)
## Data: data
##
##      AIC      BIC   logLik deviance df.resid
##    896.5    909.7   -444.2    888.5      196
##
## Random effects:
##
## Conditional model:
##  Groups   Name        Variance Std.Dev.
##  groupID  (Intercept) 10.576   3.252
##  Residual              4.081   2.020
## Number of obs: 200, groups:  groupID, 10
## Dispersion estimate for gaussian family (sigma^2): 4.08
##
## Conditional model:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  23.8343     1.3018   18.31   <2e-16 ***
## x1            1.4528     0.0764   19.02   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



Nested random effects

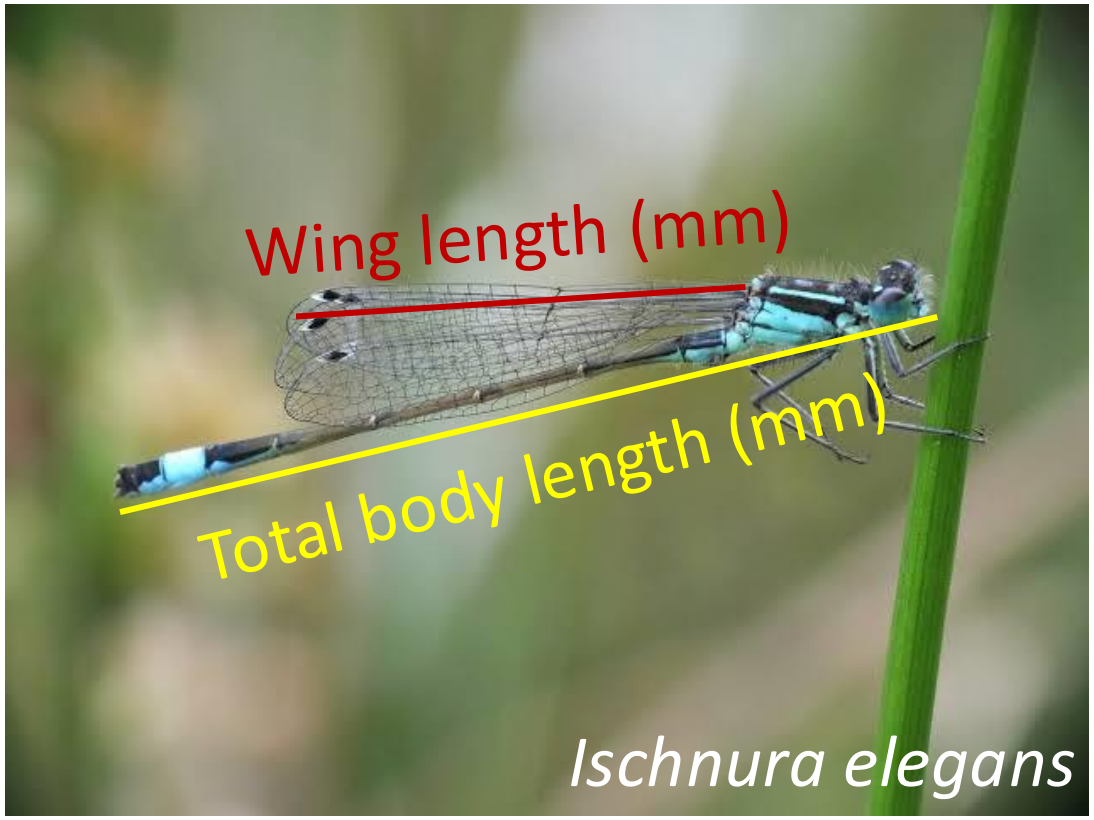
- We can fit models with several nested random effects, thus estimating hierarchical variance components
- Here there is some variance among populations, but more variance among patches within populations, and even more within patches

```
m = glmmTMB(UBW ~ 1 + (1|pop/patch), data=dat)
summary(m)
```

```
##  Family: gaussian  ( identity )
## Formula:          UBW ~ 1 + (1 | pop/patch)
## Data: dat
##
##      AIC      BIC   logLik deviance df.resid
##    980.8    994.1   -486.4    972.8      200
##
## Random effects:
##
## Conditional model:
##  Groups    Name        Variance Std.Dev.
##  patch:pop (Intercept) 3.238    1.800
##  pop       (Intercept) 1.143    1.069
##  Residual              4.463    2.113
## Number of obs: 204, groups:  patch:pop, 96; pop, 9
##
## Dispersion estimate for gaussian family (sigma^2): 4.46
##
## Conditional model:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  18.9254     0.4467   42.36   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

An example: wing-body allometry in a damselfly

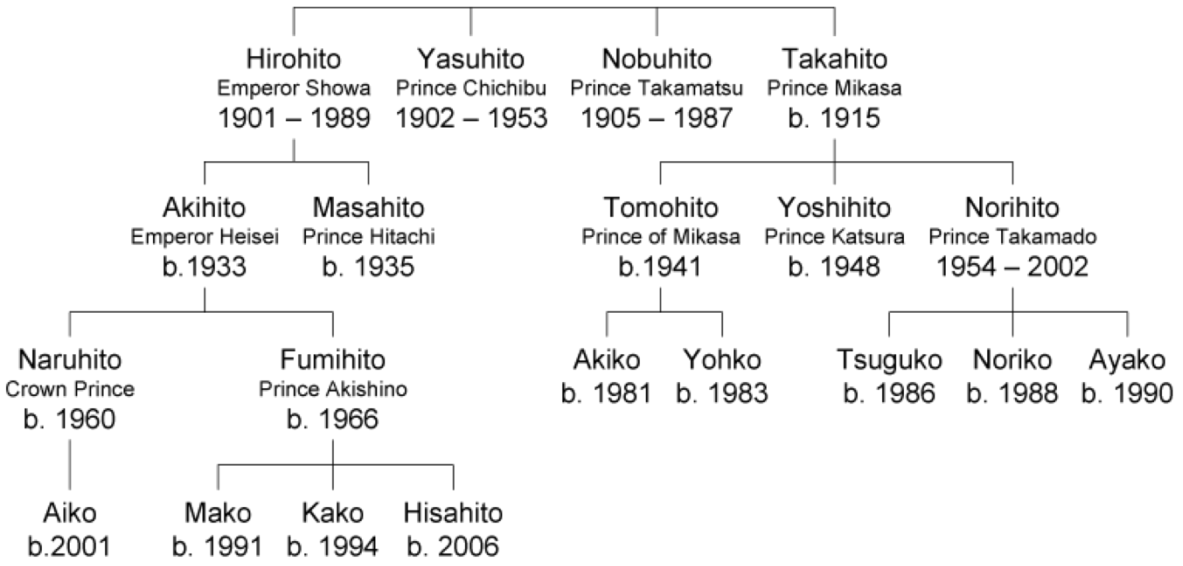
- Wing length (mm) and body length (mm) were measured in 2 species *Ischnura elegans* and *Enallagma cyathigerum*
- Samples were collected over 2 years (2020, 2021) across ~ 20 populations



More complicated nesting regime

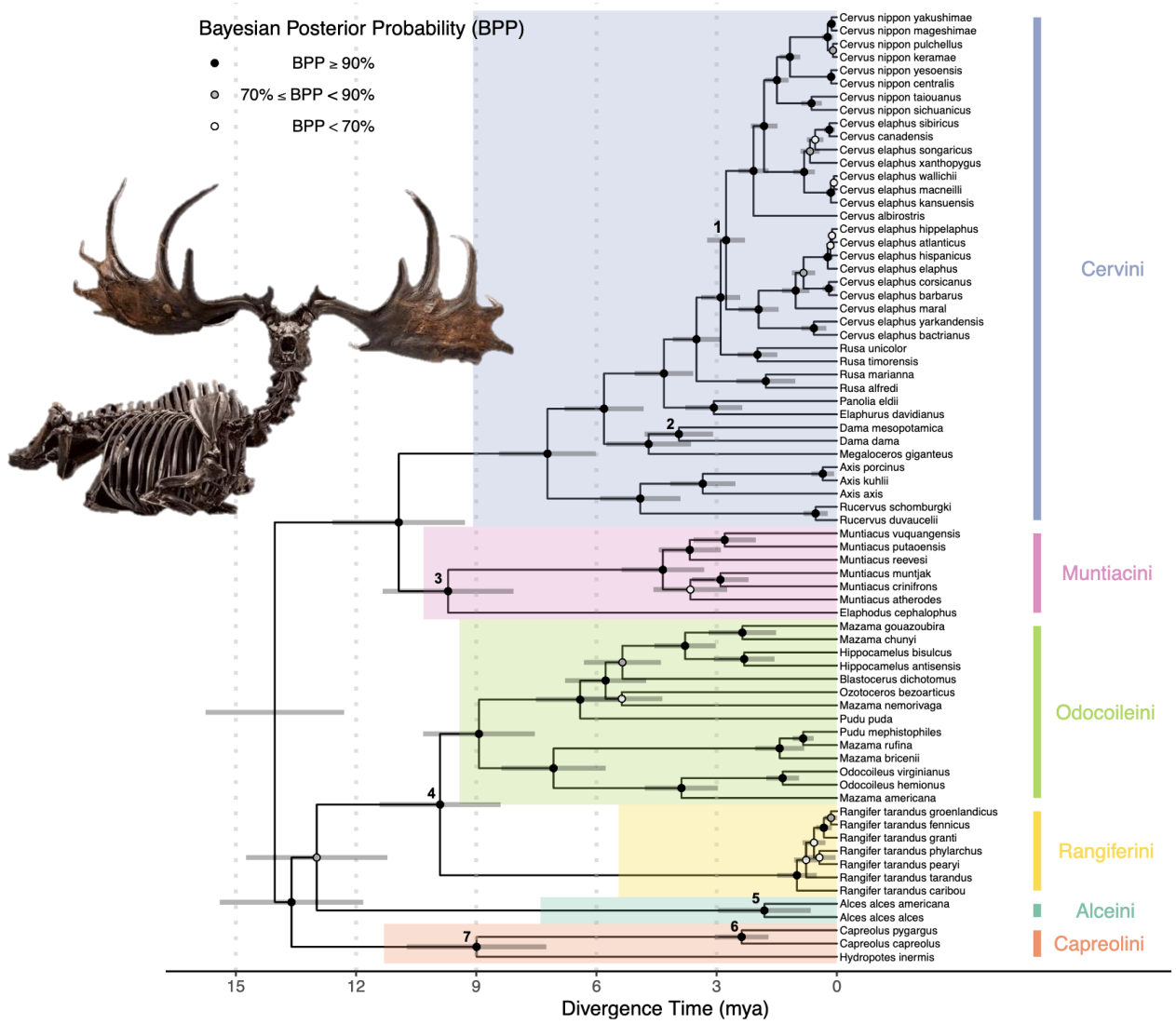
- Expected amount of shared genome (i.e., pedigree)
- A mixed effect model with pedigree as one of the random effect is called *Animal Model*
- Estimates variance component attributable to genetic differences among individuals





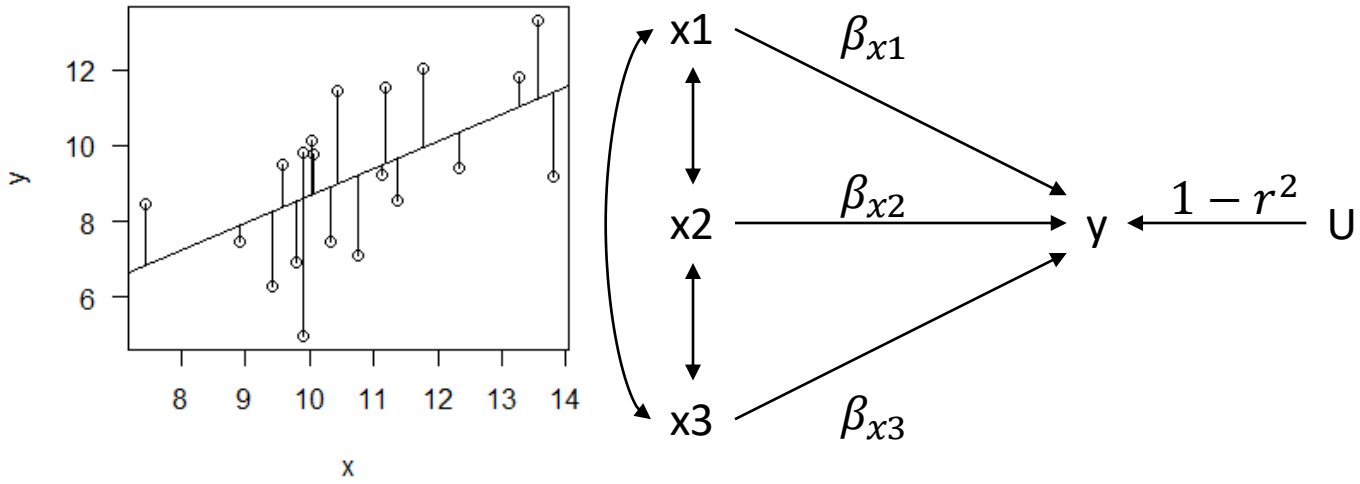
More complicated nesting regime

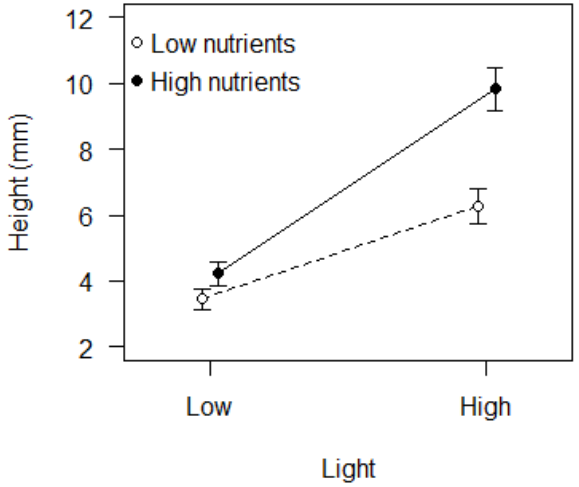
- Phylogeny (expected amount of shared history)
- A mixed effect model with phylogeny as one of the random effect is called *Phylogenetic Mixed Model*
- Estimates variance component attributable to phylogenetic differences among species (= phylogenetic signal)

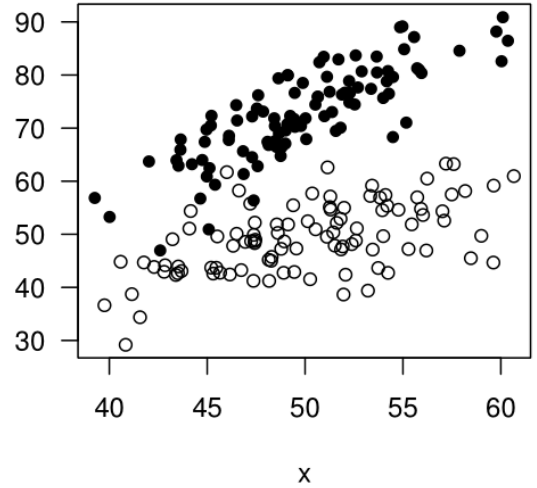


Overview of (generalized) linear (mixed) models

- Continuous covariates: (multiple) regression
- Categorical covariates: N-way ANOVA
- Continuous and categorical covariates: ANCOVA







Overview of (generalized) linear models

- Binary/proportional data: Logistic regression
- Count data: Poisson GLM
- Overdispersed count data: Negative binomial GLM

